

11.2.1 Computing the SVD from the Spectral Decomposition

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fit width



Let's see if we can turn the discussion from [Subsection 11.1.1](#) around: Given the Spectral Decomposition of $A^H A$, how can we extract the SVD of A ?

Homework 11.2.1.1. Let $A \in \mathbb{C}^{m \times m}$ be nonsingular and $A^H A = Q D Q^H$, the Spectral Decomposition of $A^H A$. Give a formula for U , V , and Σ so that $A = U \Sigma V^H$ is the SVD of A . (Notice that A is square.)

▼ [Solution](#)

Since A is nonsingular, so is $A^H A$ and hence D has positive real values on its diagonal. If we take $V = Q$ and $\Sigma = D^{1/2}$ then

$$A = U \Sigma V^H = U D^{1/2} Q^H.$$

This suggests that we choose

$$U = A V \Sigma^{-1} = A Q D^{-1/2}.$$

We can easily verify that U is unitary:

$$\begin{aligned} U^H U &= \\ &= (A Q D^{-1/2})^H (A Q D^{-1/2}) \\ &= D^{-1/2} Q^H A^H A Q D^{-1/2} \\ &= D^{-1/2} D D^{-1/2} \\ &= I. \end{aligned}$$

The final detail is that the Spectral Decomposition does not require the diagonal elements of D to be ordered from largest to smallest. This can be easily fixed by permuting the columns of Q and, correspondingly, the diagonal elements of D .

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Not all matrices are square and nonsingular. In particular, we are typically interested in the SVD of matrices where $m > n$. Let's examine how to extract the SVD from the Spectral Decomposition of $A^H A$ for such matrices.

Homework 11.2.1.2. Let $A \in \mathbb{C}^{m \times n}$ have full column rank and let $A^H A = Q D Q^H$, the Spectral Decomposition of $A^H A$. Give a formula for the Reduced SVD of A .

▼ [Solution](#)

We notice that if A has full column rank, then its Reduced Singular Value Decomposition is given by $A = U_L \Sigma V^H$, where $U_L \in \mathbb{C}^{m \times n}$, $\Sigma \in \mathbb{R}^{n \times n}$, and $V \in \mathbb{C}^{n \times n}$. Importantly, $A^H A$ is nonsingular, and D has positive real values on its diagonal. If we take $V = Q$ and $\Sigma = D^{1/2}$ then

$$A = U_L \Sigma V^H = U_L D^{1/2} Q^H.$$

This suggests that we choose

$$U_L = A V \Sigma^{-1} = A Q D^{-1/2},$$

where, clearly, $\Sigma = D^{1/2}$ is nonsingular. We can easily verify that U_L has orthonormal columns:

$$\begin{aligned} U_L^H U_L &= \\ &= (A Q D^{-1/2})^H (A Q D^{-1/2}) \\ &= D^{-1/2} Q^H A^H A Q D^{-1/2} \\ &= D^{-1/2} D D^{-1/2} \\ &= I. \end{aligned}$$

As before, the final detail is that the Spectral Decomposition does not require the diagonal elements of D to be ordered from largest to smallest. This can be easily fixed by permuting the columns of Q and, correspondingly, the diagonal elements of D .

The last two homeworks gives us a first glimpse at a practical procedure for computing the (Reduced) SVD from the Spectral Decomposition, for the simpler case where A has full column rank.

- Form $B = A^H A$.
- Compute the Spectral Decomposition $B = Q D Q^H$ via, for example, the QR algorithm.
- Permute the columns of Q and diagonal elements of D so that the diagonal elements are ordered from largest to smallest. If P is the permutation matrix such that $P D P^T$ reorders the diagonal of D appropriately, then

$$\begin{aligned} A^H A &= \text{< Spectral Decomposition >} \\ Q D Q^H &= \text{< insert identities >} \\ = Q \underbrace{P^T P}_I D \underbrace{P^T P}_I Q^H &= \text{< associativity >} \\ (Q P^T)(P D P^T)(P Q^H) &= \text{< } (BC)^H = C^H B^H H \text{ >} \\ (Q P^T)(P D P^T)(Q P^T)^H & \end{aligned}$$

- Let $V = Q P^T$, $\Sigma = (P D P^T)^{1/2}$ (which is diagonal), and $U_L = A V \Sigma^{-1}$.

With these insights, we find the Reduced SVD of a matrix with linearly independent columns. If in addition A is square (and hence nonsingular), then $U = U_L$ and $A = U \Sigma V^H$ is its SVD.

x Let us now treat the problem in full generality.

Homework 11.2.1.3. Let $A \in \mathbb{C}^{m \times n}$ be of rank r and

$$A^H A = \left(\begin{array}{c|c} Q_L & Q_R \end{array} \right) \left(\begin{array}{c|c} D_{TL} & 0_{r \times (n-r)} \\ \hline 0_{(n-r) \times r} & 0_{(n-r) \times (n-r)} \end{array} \right) \left(\begin{array}{c|c} Q_L & Q_R \end{array} \right)^H$$

be the Spectral Decomposition of $A^H A$, where $Q_L \in \mathbb{C}^{n \times r}$ and, for simplicity, we assume the diagonal elements of D_{TL} are ordered from largest to smallest. Give a formula for the Reduced SVD of A .

▼ [Solution](#)

The Reduced SVD of A is given by $A = U_L \Sigma_{TL} V_L^H$, where Σ_{TL} is $r \times r$ is diagonal with positive real values along its diagonal, ordered from largest to smallest. If we take $V_L = Q_L$ and $\Sigma_{TL} = D_{TL}^{1/2}$ then

$$A = U_L \Sigma_{TL} V_L^H = U_L D^{1/2} Q_L^H.$$

This suggests that we choose

$$U_L = A V_L \Sigma_{TL}^{-1} = A Q_L D_{TL}^{-1/2}.$$

We can easily verify that U_L has orthonormal columns:

$$\begin{aligned}
& U_L^H U_L \\
&= \\
& (A Q_L D_{TL}^{-1/2})^H (A Q_L D_{TL}^{-1/2}) \\
&= \\
& D_{TL}^{-1/2} Q_L^H A^H A Q_L D_{TL}^{-1/2} \\
&= \\
& D_{TL}^{-1/2} Q_L^H \left(\begin{array}{c|c} Q_L & Q_R \end{array} \right) \left(\begin{array}{c|c} D_{TL} & 0_{r \times (n-r)} \\ \hline 0_{(n-r) \times r} & 0_{(n-r) \times (n-r)} \end{array} \right) \left(\begin{array}{c|c} Q_L & Q_R \end{array} \right)^H Q_L D_{TL}^{-1/2} \\
&= \\
& D_{TL}^{-1/2} \left(\begin{array}{c|c} I & 0 \end{array} \right) \left(\begin{array}{c|c} D_{TL} & 0_{r \times (n-r)} \\ \hline 0_{(n-r) \times r} & 0_{(n-r) \times (n-r)} \end{array} \right) \left(\begin{array}{c|c} I & 0 \end{array} \right)^H D_{TL}^{-1/2} \\
&= \\
& D_{TL}^{-1/2} D_{TL} D_{TL}^{-1/2} \\
&= \\
& I.
\end{aligned}$$

Although the discussed approaches give us a means by which to compute the (Reduced) SVD that is mathematically sound, the Achilles heel of these is that it hinges on forming $A^H A$. While beyond the scope of this course, the conditioning of computing a Spectral Decomposition of a Hermitian matrix is dictated by the condition number of the matrix, much like solving a linear system is. We recall from [Subsection 4.2.5](#) that we avoid using the Method of Normal Equations to solve the linear least squares problem when a matrix is ill-conditioned. Similarly, we try to avoid computing the SVD from $A^H A$. The problem here is even more acute: it is often the case that A is (nearly) rank deficient (for example, in situations where we desire a low rank approximation of a given matrix) and hence it is frequently the case that the condition number of A is very unfavorable. The question thus becomes, how can we avoid computing $A^H A$ while still benefiting from the insights in this unit?

Homework 11.2.1.4. Compute the SVD of

$$A = \begin{pmatrix} \sqrt{2} & 1 \\ 0 & \sqrt{2} \end{pmatrix}.$$

► [Hint](#) ► [Solution](#)